

3753. Total Waviness of Numbers in Range II

EASY SOLUTION | Sliding Window + Clever Math Tricks | Combinatorics

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Jun LeetCode Challenge

Math

Sliding Window

Combinatorics

```
class Solution {
    static final int[] waves = new int[570];

    static {
        int j = 0;
        for (int i = 0; i < 1000; i++) {
            int r = i % 10;
            int m = (i / 10) % 10;
            int l = (i / 100) % 10;
            if ((m > Math.max(l, r)) | (m < Math.min(l, r)))
                waves[j++] = i;
        }
    }

    public long totalWaviness(long A, long B) {
        return waveCount(B) - waveCount(A - 1);
    }

    private long waveCount(long num) {
        if (num < 100) return 0;
        long res = 0;
        for (int p : waves)
            res += countWays(num, p);
        return res;
    }

    private long countWays(long num, int pattern) {
        long type = pattern < 100 ? 1 : 0;
        long count = 0, mult = 1;

        for (int i = 0; mult * 100 <= num; i++) {
            long pre = num / (mult * 1000);
            long cur = (num / mult) % 1000;
            long suf = num % mult;
            long ways = 0;

            if (cur > pattern)
```

```

        ways = pre - type + 1;
    else if (cur == pattern) {
        ways = Math.max(0L, pre - type);
        count += suf + 1;
    } else
        ways = Math.max(0L, pre - type);

    count += ways * mult;
    mult *= 10;
}

return count;
}
}

```

Phase 1 : Base Pattern Enumeration

The waviness is determined by a 3-digit sequence, so it is much more efficient if we enumerate / precompute all these base patterns:

There only 570 of them:

peak			valley		
possible left	mid	possible right	possible left	mid	possible right
5		5	9		9
4		4	8		8
3	5	3	7	5	7
2		2	6		6
1		1	5		5
0		0	4		4

Peaks:

$$\text{mid}^2 = 0^2 + 1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 + 7^2 + 8^2 + 9^2 = 285$$

Valleys:

$$(9 - \text{mid})^2 = 9^2 + 8^2 + 7^2 + 6^2 + 5^2 + 4^2 + 3^2 + 2^2 + 1^2 + 0^2 = 285$$

$$\text{Peaks} + \text{Valleys} = 570$$

BASE PATTERNS

Type 0: Prefix is a non-zero:

101,102,103,104,... 967,968,969,978,979,989

Type 1: Prefix is a zero:

010,020,021,030 ... 093,094,095,096,097,098

If a number n contains one of this pattern, it is only valid if it has a non-zero leading prefix:

Ex: 58734093...

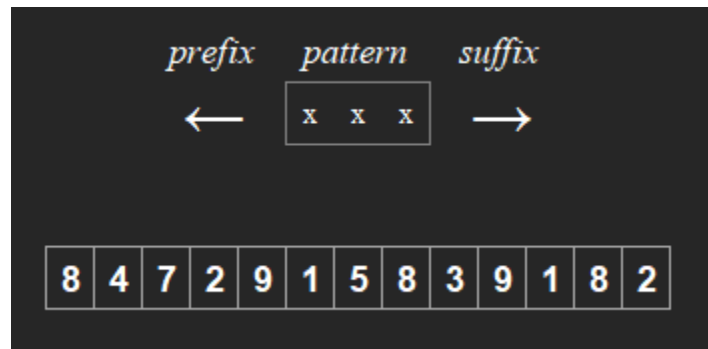
58734093... → prefix

Otherwise, we restrict its contribution.

Phase 2: Sliding-Window × Base Patterns

Consider an arbitrary number:

$n=8472915839182$



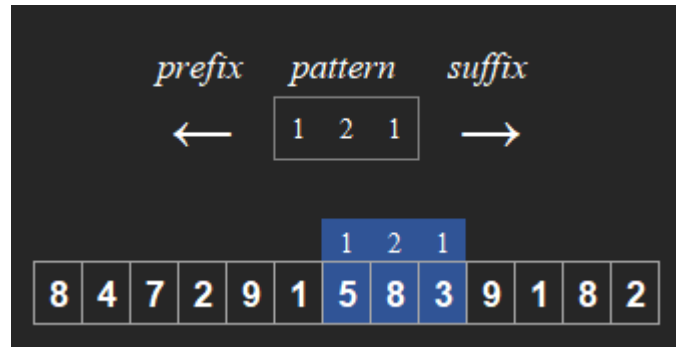
We perform a sliding window scan across the digits of n while evaluating against all 570 valid base patterns.

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Phase 3: Find the number of Combinations

We take a snapshot state of the scan:



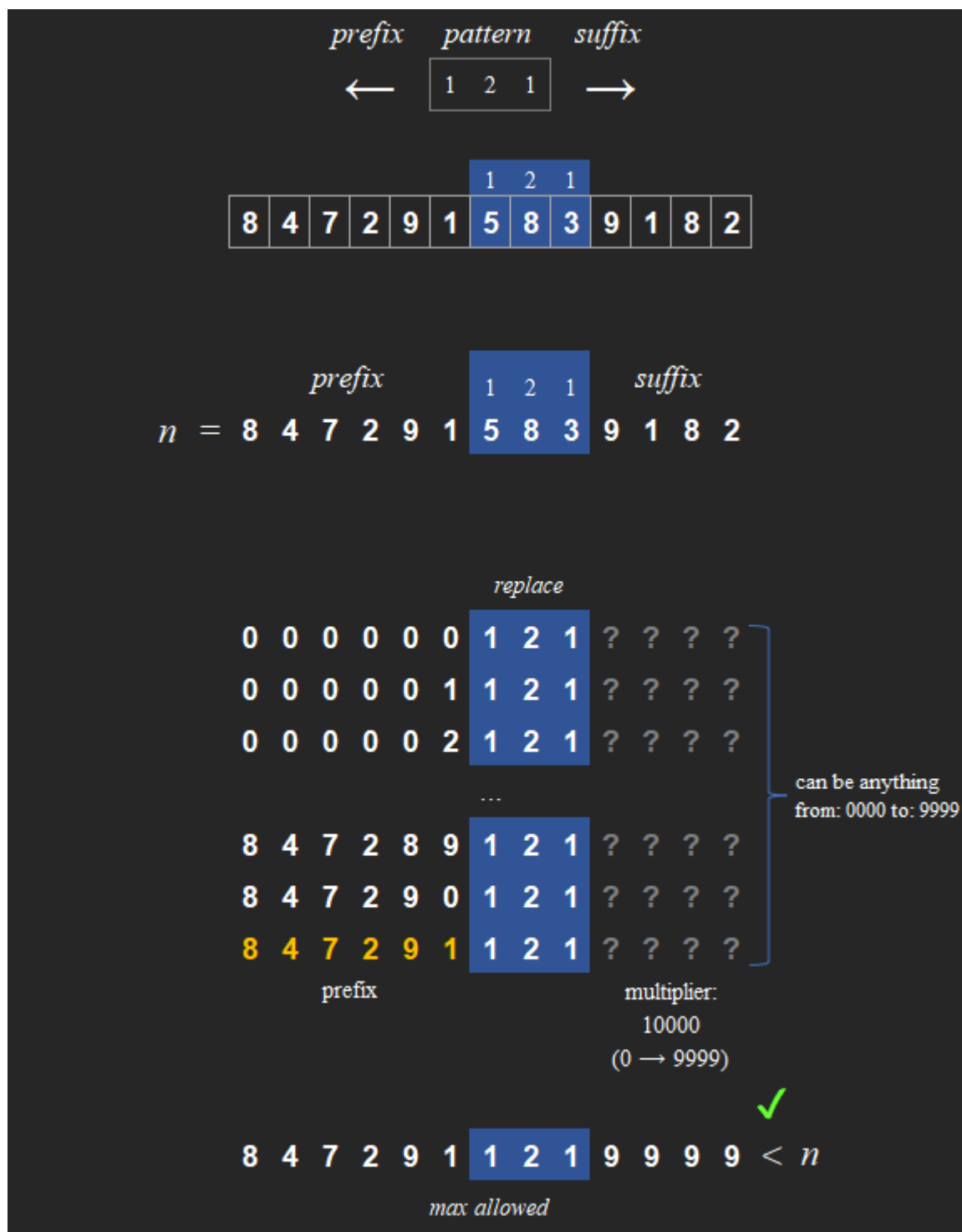
Take the current window curr, and compare this against the pattern:

There are 3 possible cases:

CASE 1: $\text{curr} > \text{pattern}$

In this state, we have $\text{pattern}=121$ and current window $\text{curr}=583$

Without manually scanning from 0 to n, we can determine the exact count of numbers with 121 at that specific window:



The formula to find the number of combinations (ways):

$$ways = \underbrace{(prefix + 1)}_{0 \rightarrow prefix} \times multiplier$$

$$ways = (847291 + 1) \times 10000$$

$$= 8\,472\,920\,000$$

$$\text{ways} = (\text{prefix} + 1 - \text{type}) \times \text{multiplier}$$

In this state, we have pattern=583 and current window curr=583



$$ways = \underbrace{(prefix)}_{0 \rightarrow prefix-1} \times multiplier + \underbrace{(suffix + 1)}_{0 \rightarrow suffix}$$

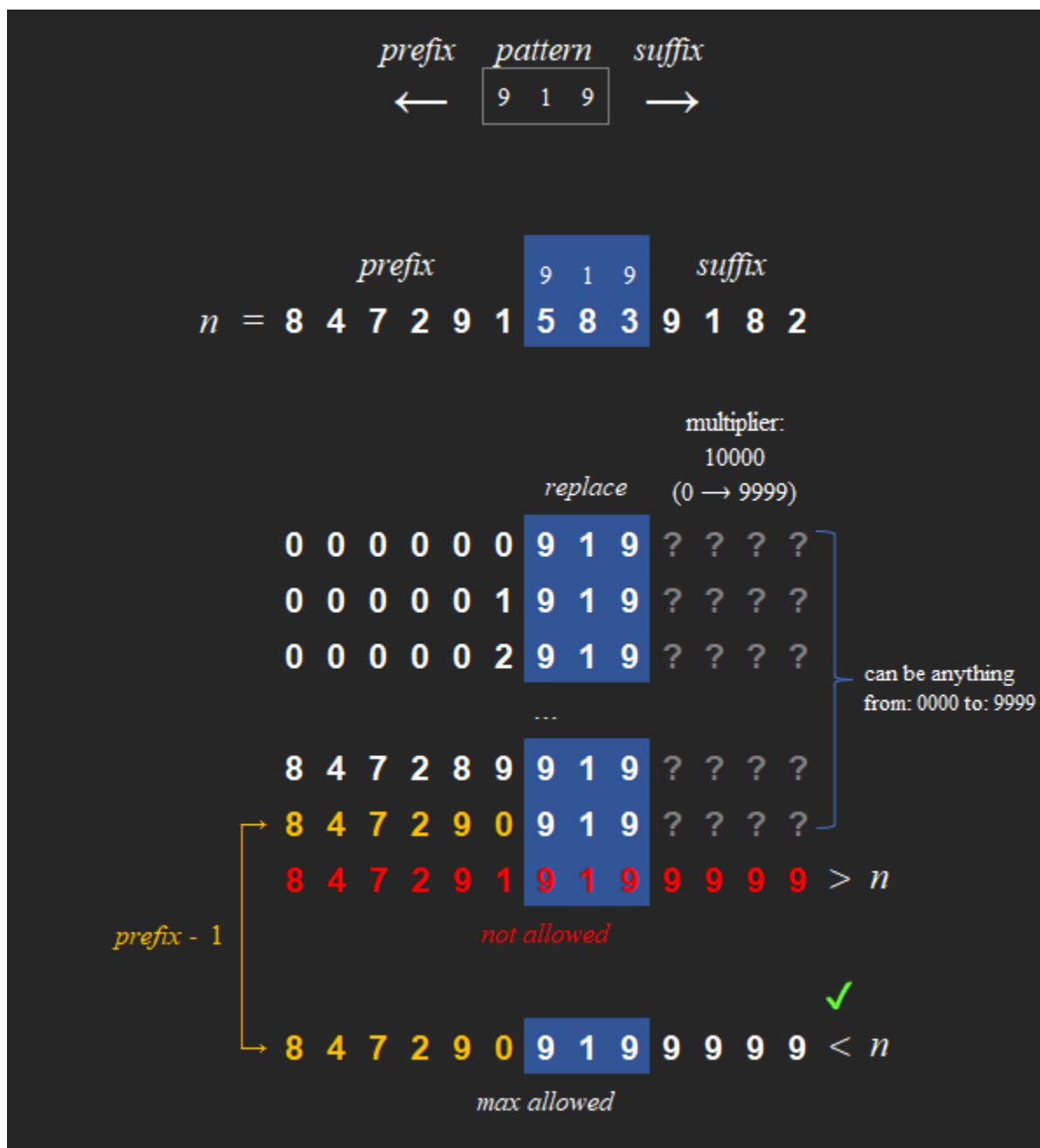
$$\begin{aligned} ways &= 847291 \times 10000 + (9182 + 1) \\ &= 8\,472\,910\,000 + 9183 \\ &= 8\,472\,919\,183 \end{aligned}$$

Adjust the formula since some zero-prefix patterns can be invalid:

$$ways = (\max(0, prefix - type) \times multiplier) + (suffix + 1)$$

CASE 3: $curr < pattern$

In this state, assume we have a $pattern=919$ and the current window is $curr=583$.



The formula to find the number of combinations (ways):

$$ways = \underbrace{(prefix)}_{0 \rightarrow prefix-1} \times multiplier$$

$$ways = 847291 \times 10000$$

$$= 8\ 472\ 910\ 000$$

Adjust the formula since some zero-prefix patterns can be invalid:

$\text{ways} = \max(0, \text{prefix-type}) \times \text{multiplier}$

For any given window, the exact *combinations* are determined by comparing the current window segment against the target *pattern*.

We accumulate up all these *combinations* against all *patterns* exhaustively until we finished scanning the entire number:

Finally, perform *inclusion-exclusion* principle to find the exact *totalWaviness* between the given range.

$\text{return waveCount(B)} - \text{waveCount(A-1)}$.

Time Complexity: $O(P \times \log n)$

Space Complexity: $O(1)$